CricMind

Cricket intelligence where every number is checked before you see it

Bangladesh AI Business Summit — Primary Selection

Section A: Project Idea Submission

Section B: Project Supporting Document

Category:	AI for sports analytics / decision support (B2B)
Research basis:	*CricketMind: A Verification-Gated Multi-Agent Architecture for Multi-Stakeholder Cricket Analytics*
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Document A: Project Idea Submission

1 Executive Summary

Cricket decisions in Bangladesh, who to buy, how to plan against a bowler, whether a player is fit for a final, are often made from scattered spreadsheets, memory and gut feeling. AI assistants could help, but a chatbot that sounds confident and quietly invents a strike rate is dangerous when money and careers depend on it.

CricMind is an AI analyst that answers cricket questions in plain language and tailors the answer to the asker: a coach, franchise owner, scout, medical team, academy or broadcaster. Its key idea, shown in the CricMind research, is a strict separation of roles. Only two audited calculation engines may produce numbers. The AI agents write the explanation, and a "Faith Check" gate compares every number they cite with the engines. A wrong number is sent back for correction or removed, and the report says so.

The first product is a text-based match-preparation and scouting brief for Bangladeshi franchises, domestic teams and cricket academies. Video-based shot analysis is a later add-on. The business is a subscription with per-report and data-integration options.

2 Problem Statement

Who faces it: analysts and coaches at franchises, domestic teams and academies; franchise owners deciding on players; scouts; medical staff; broadcasters and fan platforms.

The problem has two layers.

(a) **Decision-makers need fast, tailored analysis.** Different stakeholders ask different questions of the same data: a coach wants tactics, a medical team wants workload, a franchise wants value for money.

(b) **AI tools that could help fail on trust.** A language model can produce a fluent report containing figures it never calculated. A report that sounds confident but was never checked against evidence can mislead the people who rely on it.

Current issues in Bangladesh Cricket System:

- The Bangladesh Premier League has had unstable franchise economics; reports describe franchises losing interest, payment defaults to players, and demands for revenue sharing [S1, S4].
- The BCB has since moved toward revenue sharing: 30% of net profit from media rights, ground rights and ticket sales to franchises, for a five-year cycle [S3]. When franchises share in profit, spending wisely on players and preparation matters more. [ASSUMPTION] This must be validated by interviews.

3 Proposed AI Solution

What users do. Type a question (for example, "Compare Litton Das's cover drive vs his pull against fast bowlers in the powerplay") or, in a later version, upload a short clip of a batting shot.

What the system produces. A short report for the asker's role, with each figure traceable to a calculation, plus a status line stating whether anything was withheld.

Illustrative scenario A coach asks the question above. The system finds the matching deliveries, calculates a strike rate for each shot, simulates six deliveries aimed at the weaker shot, and returns a Coaches finding and a Players finding. Before display, every cited number is checked. If an agent had cited a wrong number, only that agent would be asked to redo its answer; if it failed again, the finding would be left out and the report would say so.

4 How the AI Works

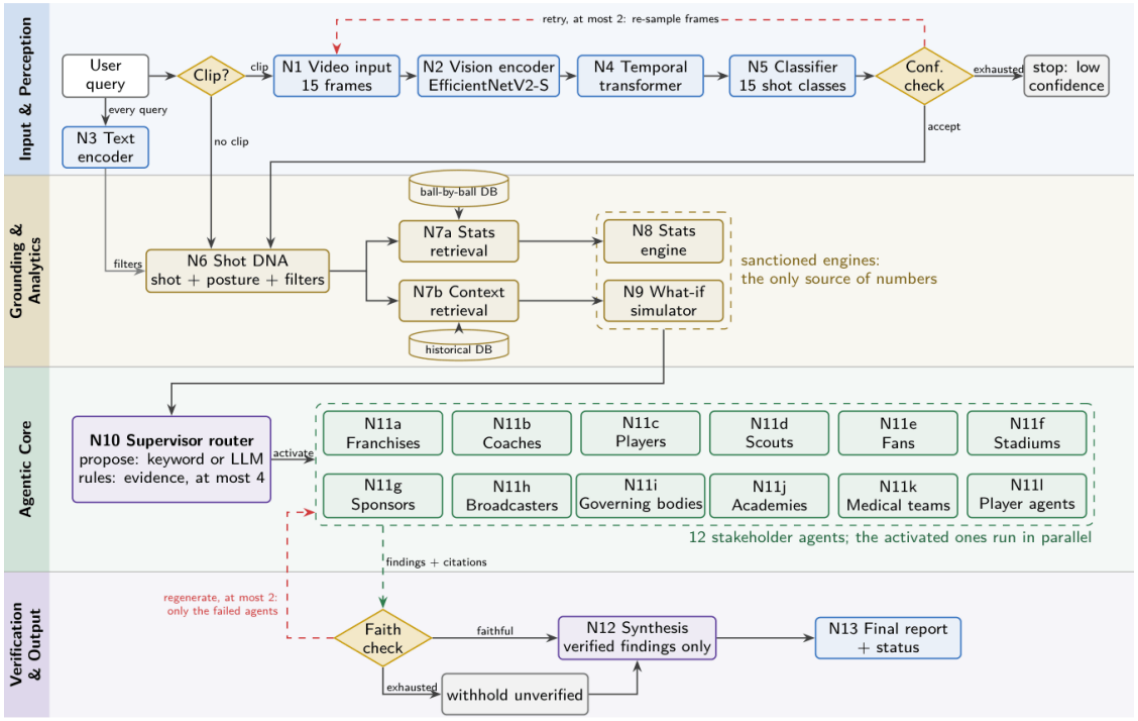


Figure 1: System architecture and workflow pipeline.

Key concepts in two layers

Concept	Explanation
Shot DNA	Structured record: accepted shot label, query filters (player, bowler type, phase, named shots) and posture measurements (weight transfer, bat-face angle, stride length). Text-only queries leave posture fields empty rather than guessed.
Perception front end	15 uniformly sampled frames; EfficientNetV2-S gives a 1280-d vector per frame; a 2-layer, 4-head temporal transformer fuses them; classifier predicts one of the 15 CricShot10k classes with a confidence score.
Sanctioned engines	N8 computes exact aggregates (strike rate, dismissal rate per shot); N9 is a Monte Carlo simulator reporting outcome probability with a 95% confidence interval.
Two retrieval indexes	N7a precise ball-by-ball index; N7b broader historical index.

Concept	Explanation
Supervisor router	Keyword or LLM router proposes; the same fixed rules then drop agents lacking evidence, cap at four, and default to Coaches+Players. LLM router: Qwen2.5-7B via Ollama, temperature 0, fixed seed.
Faith Check and citation contract	Each agent returns its answer plus a citation record. Mismatch regenerates that agent only (max 2), then withholds. **Limitation:** the gate checks citations, not prose; swapped subjects and uncited figures passed. Fix: force every number to be emitted as a subject-keyed citation (JSON schema or tool call) and render sentences from verified citations.

5 Innovation and Differentiation

Evaluation and validation profile (synthetic setup):

- Numbers-only-from-engines design with a verification gate; it rejected a fabricated citation (`best_sr` = 999.0) and regenerated only the offending agent.
- Predictable cost: $S = S_0 + 5r + 3g + d$ predicted the superstep count of all seven tested scenarios exactly. Worst case: 29 supersteps and at most 12 agent calls, fixed when the system is built ($S_0 = 12$ with a clip, 9 without; r vision retries; g regenerations; $d = 1$ if findings are withheld).
- Router comparison: on 12 plain prompts avoiding every keyword, the keyword router activated the intended agent 0/12 times, the LLM router 9/12; LLM routing was identical across 3 runs; mean routing time 3.0 s.
- A candid finding: the gate misses swapped subjects and uncited numbers, which defines exactly what the product must enforce.

Proposed product innovation (not yet built): role-tailored reports for Bangladeshi stakeholders from one engine; subject-keyed citation enforcement; text-first entry so value does not depend on video quality.

Possible future advantage (unproven): Bangladesh-specific data and Bangla reports; a data flywheel from customer match data.

Most defensible differentiator: auditability. Global vendors such as CricViz offer deep data, models and AI commentary [S6, S7]; CricMind does not claim better data. It claims that every number in a role-specific report can be traced to a calculation, which matters when reports drive spending and selection decisions.

6 Target Market and Customers

Beachhead: Bangladeshi franchise and domestic-team analysts and coaching staff, plus a small number of cricket academies.

- **Primary users:** coaches, analysts, scouts, academy trainers.
- **Paying customers:** franchise management (BPL had seven franchises in 2025 [S1, S2]), academies, later media organisations.
- **Institutional buyers:** BCB high-performance units; broadcasters and fan platforms.

- **Later sectors:** sports media, sponsorship evaluation, sports medicine.

Why start here: decisions are high-stakes and repetitive; franchises now share in profit [S3], a plausible motive to improve decisions [ASSUMPTION]; a small, reachable group allows direct pilots.

Caution: the franchise segment is small and financially fragile [S1, S4], so the plan adds academies and media rather than relying on franchises alone.

Option	Pain	Pay	Feasible	Verdict
Franchise/coach briefs	High	Med	High	Chosen (start)
Academy coaching reports	Med	Low	High	Second segment
Media / fan stats API	Med	Med	Med	Third segment
Medical workload brief	High	Low	Low*	Later

Qualitative judgements by the author, not data. *Health-data rules apply.

7 Business Model

- **Core:** software-as-a-service subscription for team analysts and academies.
- **Add-ons:** per-report pricing for one-off requests (auction week, final preparation); API access for media and fan platforms.
- **Service layer:** setup and data-onboarding fee in year one, because customer data will be messy.
- **Value claim to test (not yet proven):** less analyst time per brief and fewer errors in numbers shown to decision-makers.

8 Market Opportunity and Competition

Market evidence:

- A market-research summary quoted on an aggregator page projects the global cricket analysis software market at about USD 0.89 billion by 2029, growing about 13.6% a year [S7]. This is a rough, unverified vendor-report figure, not a Bangladesh figure.
- UNB reported BPL title sponsorship and media rights sold for under USD 5 million over three seasons, against over USD 24 million for one PSL season's broadcast rights [S4], indicating limited local budgets and the need for a low-cost product.

Type	Examples	Strength	Gap targeted
Direct	CricViz (UK; Cricket Australia partner) [S6, S7]	Deep data, models, global clients	Not Bangladesh-focused; local pricing fit not established
Direct	Other cricket-analysis vendors named in market reports [S7]	Established team/academy tools	No verified numeric reports
Indirect	Hawk-Eye, general video analysis tools	Ball tracking, tagging	Not role-tailored written briefs
Manual	Excel, scorebooks, notes	Cheap, familiar	Slow; errors untraced

Type	Examples	Strength	Gap targeted
Generic AI	General chatbots	Easy, flexible	Can state unchecked numbers

Position: not "better data" but "provable numbers, plain language, role-specific, affordable" [ASSUMPTION] until tested.

9 Technical Feasibility and Current Development Status

Completed :

- LangGraph state graph, 30 nodes, one per component.
- Deterministic stand-ins for every node except the router; no video decoded; the classifier returns a configurable label and confidence.
- Synthetic corpus: 150 deliveries and 4 historical records with known ground truth.
- Template agents emitting the same citation record an LLM agent would be required to produce.
- Execution profile of 7 scenarios; the cost model matched all 7 exactly.
- Router comparison on 29 author-written prompts, with 3 runs of the LLM router on a single consumer GPU.
- Faith Check boundary test: honest answer passes; fabricated citation rejected; swapped-subject and uncited-figure cases pass.

Not completed: real data; real video classification (accuracy outside paper scope); LLM-written agents; real retrieval; user interface; production security; any customer pilot; real latency measurement.

To reach an MVP:

- Real ball-by-ball data pipeline and licence.
- Subject-keyed citation schema enforced on LLM agents.
- Routing before analytics, so only needed statistics are computed (the paper notes unused ones are currently computed).
- Graceful handling of players with no deliveries (not yet handled).
- Simple web or mobile interface; Bangla option.

Deployment sketch: cloud or local server hosting the graph, a small delivery database, and a hosted or local LLM. Operating cost is unknown; main drivers will be LLM calls (up to $4 \times 3 = 12$ agent calls per query in the worst case), storage and data licensing. Routing took about 3.0 s on a consumer GPU in the paper; end-to-end latency is untested.

Expo demonstration: a live run of the existing prototype on our synthetic test data. We ask it a cricket question and show which stakeholder agents it activates, then plant a fabricated statistic in one agent's answer so visitors can watch the Faith Check reject it in real time, re-run only that agent, and record the outcome in the report's status line.

10 Implementation Roadmap

Phase	Focus
Months 0–3	Validate: interview 10–15 analysts, coaches, academy heads and one media contact; secure a small real dataset with permission; fix the citation schema.
Months 3–6	Build text-only MVP with real data and 3–4 agents (Coaches, Players, Scouts, Franchises); test against hand-calculated answers.
Months 6–9	Free pilots with 2–3 teams/academies; measure analyst time saved and number errors caught.
Months 9–12	Paid pilots; add Bangla output; decide on video.
Year 2+	Video path (after proper accuracy evaluation), more agents, media API, regional expansion.

Timings are planning assumptions, not commitments.

11 Risks and Mitigation

Risk	Mitigation
No access to real data	Start with academy/franchise partners; budget for licensing; consider self-collection.
Small local paying market	Multi-segment plan; low-cost text-first MVP; regional expansion later.
Customer payment problems (franchise instability [S1, S4])	Annual prepayment; include academies and media.
LLM errors beyond the gate (swapped subject, uncited figure)	Enforce subject-keyed citations; render text from verified facts; block uncited numbers.
Perception unproven	Do not sell video until accuracy is measured on real clips.
Competitor strength	Position on auditability and local fit, not data depth.
Privacy and regulation	Comply with the Personal Data Protection Ordinance 2025 [S8, S9]; avoid personal health data in the MVP; track the National AI Policy [S10, S11].
Trust and adoption	Show the source of every figure; keep human decision-makers in charge.

Document B: Project Supporting Document

1 Research Foundation

What we investigated. Whether an agentic system can serve multiple cricket stakeholders from one pipeline while guaranteeing that cited numbers come from trusted engines.

Objectives :

1. Accept a shot clip or a text-only question.
2. Provide a structured Shot DNA and two sanctioned engines.
3. Route to a relevant subset (at most 4 of 12) of stakeholder agents.
4. Verify cited figures with a faithfulness gate and bounded regeneration.
5. Characterise cost and compare keyword and LLM routers.

Prior work and gap. The paper reviews trajectory forecasting (Yeh et al.[S13]; Teoh [S13]), action spotting (Ochin et al. [S13]), simulation environments (Kurach et al. [S13]), robotic-soccer reinforcement learning (Brandão et al. [S13]), supply-chain agents (Gupta [S13]; Hazenberg et al. [S13]) and FanCric for fantasy cricket (Bhatnagar [S13]). Its claimed gap, to the authors' knowledge: none combines video-grounded shot classification, retrieval-backed statistics, dynamic multi-stakeholder routing and a faithfulness gate in one cricket system.

Link to the product. The product keeps the verified-numbers backbone and replaces stand-ins with real data and LLM agents.

2 Proposed System and Technical Methodology

Data. Synthetic corpus (150 deliveries, 4 historical records). Product: ball-by-ball data and historical records;

Models and algorithms.

- Vision: EfficientNetV2-S (1280-d per frame), 15 frames.
- Temporal: 2-layer, 4-head transformer.
- Classifier: 15 CricShot10k classes [S13]; confidence gate at 0.70; at most 2 re-samples.
- Statistics engine: exact aggregates. What-if simulator: Monte Carlo with 95% confidence interval.
- Router: keyword scoring plus structural signals, or Qwen2.5-7B LLM.
- Faith Check: citation-to-engine comparison.

Architecture. 13 conceptual nodes (N1–N13), 30 nodes in the LangGraph implementation, 12 stakeholder agents (franchises, coaches, players, scouts, fans, stadiums, sponsors, broadcasters, governing bodies, academies, medical teams, player agents).

Outputs. A report plus a status: `ok`, `low_confidence` or `unverified_claims_dropped`.

Evaluation methods in the paper. Superstep and node-run counting from the framework debug stream; closed-form cost model; paraphrased-prompt routing test; adversarial agent answers for the gate.

3 Research Results and Evidence

3.1 Execution profile

Evaluation query, each scenario run once; node bodies are deterministic.

Scenario	r	g	Supersteps	Node runs	Agent calls
Video, confident	0	0	12	16	2
Text-only	0	0	9	11	1
Video, 1 retry	1	0	17	21	2
Video, 1 fabrication	0	1	15	19	3
Text-only, 1 fabrication	0	1	12	14	2
Retries exhausted	2	0	15	16	0
Worst case	2	2	29	35	6

Cost model: $S = S_0 + 5r + 3g + d$; degraded case $S = 4 + 5r + 1$. It matched all seven scenarios. Worst-case bound $12 + 10 + 6 + 1 = 29$ supersteps; agent-call bound $4 \times (1 + 2) = 12$ (not reached in runs).

3.2 Routing comparison (29 prompts)

Measure	Keyword	LLM (Qwen2.5-7B)
Keyword prompts: target activated	12/12	12/12
Plain prompts: target activated	0/12	9/12
Fell back to default agents	14/29	2/29
Agents activated per prompt (mean)	1.79	1.41
Identical in all 3 runs	—	29/29
Routing time per prompt (mean)	—	3.0 s (2.8–3.6 s)

3.3 Faith Check boundary test

- Honest answer: passes.
- Fabricated citation (`best_sr` = 999.0): rejected, regenerated once.
- Swapped subjects with identical cited values: **passes** (not caught).
- Extra uncited figure: **passes** (not caught).

3.4 What the findings establish, and what they do not

Establish: the control flow behaves as designed, with predictable bounded cost; LLM routing copes with paraphrased questions far better than keywords on this small set, with stable output at temperature 0; the gate blocks fabricated citations.

Do not establish: quality of analysis, usefulness to real users, or accuracy on real data (components are stand-ins); shot-classification accuracy; real latency; generalisation of routing (29 author-written prompts, 12 plain, one machine; the paper calls it indicative); protection against wrong-subject or uncited figures without the citation schema.

Remaining validation: real-data accuracy tests; classifier evaluation; larger independent routing set; user trials.

4 Product Design and Use Cases

Core MVP features:

1. Plain-language question box (English first, Bangla output next).
2. Role selector or automatic routing to up to 4 agents.
3. Report with a "how this number was calculated" link for every figure.
4. Status banner: complete / some findings withheld.
5. Exportable PDF brief.

Example user journey (franchise analyst). Ask → system extracts player, bowler type, phase → retrieves deliveries → engines calculate → Coaches and Scouts agents write → Faith Check → report with traceable figures → analyst edits and shares with the coach.

Use case 1, match-up brief: a coach compares two shots against pace in the powerplay before a game.
Use case 2, academy report: an academy head asks how a young batter's stroke selection compares across phases (video path later).

MVP scope: text-only; 3–4 agents; real data from one or two partners; no video; no personal medical data.

5 Limitations and Future Development

Limitations: Stand-ins on a synthetic corpus; perception accuracy out of scope; routing set of 29 prompts is indicative; stability measured on one machine; the gate guarantees only what the citation contract enforces; no handling yet for a player with no deliveries; statistics are computed before routing, so some go unused.

Future development : Subject-keyed LLM agents; real shot-annotated ball-by-ball data; counterfactual questions such as a changed batting order; routing before analytics; giving the LLM router an evidence summary; measuring the cost of containment as fabrication rises.

6 Conclusion

CricMind demonstrated, in a controlled prototype, that a multi-stakeholder cricket assistant can have bounded cost, routing that copes with plain language, and a gate that blocks fabricated figures, while openly showing where that gate stops. CricMind proposes turning this into a focused, text-first brief writer for Bangladeshi teams and academies, with the first year spent on data, citation enforcement and real customer validation.

7 References and Supporting Evidence

References

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